Response to Reviewer

Manuscript Title: Enhancing Statistical Accuracy in Gene Perturbation Studies
Manuscript ID: BIOSYS-D-25-00695
Author: Vijender Kalmotia

Dear Prof. Igamberdiev,
Thank you for the opportunity to revise and resubmit our manuscript entitled "Enhancing Statistical Accuracy in Gene Perturbation Studies" (BIOSYS-D-25-00695R1). We sincerely thank you and the reviewers for their valuable feedback. We have carefully revised the manuscript to address all concerns raised. Below we provide a detailed response.

Response to Reviewer #1

We thank the reviewer for the positive evaluation.

Comment:
"My comments have been addressed. I suggest open the source code and data."

Response:
We appreciate this suggestion. To ensure transparency and reproducibility, we have provided the full implementation of our analysis pipeline as **supplementary material (ZIP file)** with this submission.

Response to Reviewer #2

We thank the reviewer for their careful evaluation and important concerns regarding originality and clarity.

Comment 1:

Response:
We acknowledge the reviewer's concern and have made substantial revisions to clarify the relationship between our work and the original SCEPTRE framework.

- We explicitly state throughout the manuscript that **SCEPTRE was originally introduced by Barry et al. (2021)**.
- All language suggesting that SCEPTRE is newly proposed in this work has been removed.
- A dedicated section titled **"Relationship to Prior Work"** has been added.

Importantly, we clarify that this work does **not introduce a new statistical framework**, but instead provides:

- an **extension of SCEPTRE to high-MOI CRISPR screening settings**,
- a **scalable computational implementation**, and
- a **systematic empirical and biological evaluation**.

We also explicitly outline novel contributions, including:

- adaptation to high-MOI conditions,
- regression distillation for computational efficiency,
- skew-t null approximation for improved p-value estimation, and
- an integrated biological validation pipeline.

Comment 2:

Response:
We have added a detailed workflow description and structured pipeline in the Methods section, outlining:

- preprocessing,
- conditional randomization testing,
- resampling procedure,
- statistical inference, and
- downstream validation.

This improves transparency and interpretability of the methodology.

Comment 3:

Response:
All figure captions have been revised to include complete descriptions of:

- datasets used,
- methods applied, and
- key results.

Additional Improvements

- Expanded discussion on limitations, including independence assumptions and computational cost.
- Added sensitivity analysis and improved presentation of results.
- Improved clarity and consistency throughout the manuscript.

We believe these revisions substantially improve the manuscript and address all reviewer concerns. We hope the revised version is now suitable for publication.

Thank you for your consideration.

Sincerely,
Vijender Kalmotia